

The “Manual to Digital” Migration Plan

With South Africa’s current medical file system where hospitals and clinics use paper files for patient records, this system is slow, files might get lost or damaged or even worse a patient might die due to lack of information and possible allergies to certain medications.

This project focuses on replacing the current paper-based record used by local medical clinics and hospitals with a digital system. The purpose of this project is to design and implement a digital system that allows patient data to be stored securely using a database and accessed quickly by staff, this will improve the service delivery, and patient safety as it will reduce errors, improve operational efficiency and overall quality of healthcare services

The objectives of the project are:

- To ensure data security and confidentiality of patient records
- To create a process flow for the patient movement
- To store patient data using SQL and DBMS

1. Project scope

Included:

- Designing and developing process flow diagrams
- Designing and developing database system
- Data capturing and managing using SQL
- Testing the database system for accuracy and usability

Excluded:

- Hardware installation, e.g. servers, computers)
- Internet or network infrastructure upgrades
- External system integrations - national health databases

Scope boundary:

The project focuses only on digitising patient records and workflow within clinics and hospitals, it does not extend to external systems, national databases or infrastructure improvement. The focus is strictly on digital record management and workflow optimisation with the medical field.

2. System Requirements

The following system requirements were identified to ensure that the new digital clinic system meets the needs of the clinic staff and improves not only time but also the management of patient records

1. Functional Requirements

- The receptionist must be able to capture the patient's details such as ID, date of birth, full name, surname, gender, phone number, address, next of kin details and the medical history.
- The system must store patient details in a database management system(DBMS)
- The system must allow the staff to search for a patient record quickly using the patient details such as Id, name or phone number
- The system must allow the receptionist to register new patients
- The system must allow existing patient records to be updated when details change
- The system must allow the doctor to view the patient's medical record before consultations
- The system must allow the doctor to add consultation notes, diagnosis, treatment, and prescriptions to the patient record.
- The system must record the patient flow from reception to consultation.
- The system must allow authorized users to delete incorrect or duplicate records when necessary.
- The system must generate simple reports, such as the number of patients seen per day.
- The system must save all the entered data using SQL commands into the database
- The system must provide login access for authorized users such as receptionists, nurses, and doctors.

2. Non-Functional Requirements

- The system must be easy to use so that clinic staff can learn it quickly.
- The system must be fast and accurate when searching and retrieving the patient records.
- The system must be secure so that only authorized staff can access patient information.
- The system must be reliable and available during clinic working hours.
- The system must be able to send file records to other clinics/hospitals in case of referrals.
- The system must keep patient information confidential.
- The system must have backup features to prevent data loss.
- The system should be able to run on the clinic's available computers.

3. Hardware Requirements

- Desktop computers or laptops for reception and doctors office
- Printer for printing reports or prescriptions
- Server or main computer for hosting the database
- Stable power supply

4. Software Requirements

- Operating system such as windows or Linux
- DBMS such as MySQL or PostgreSQL
- SQL for capturing, updating, and retrieving data
- Application software for the clinic system
- Antivirus and backup software

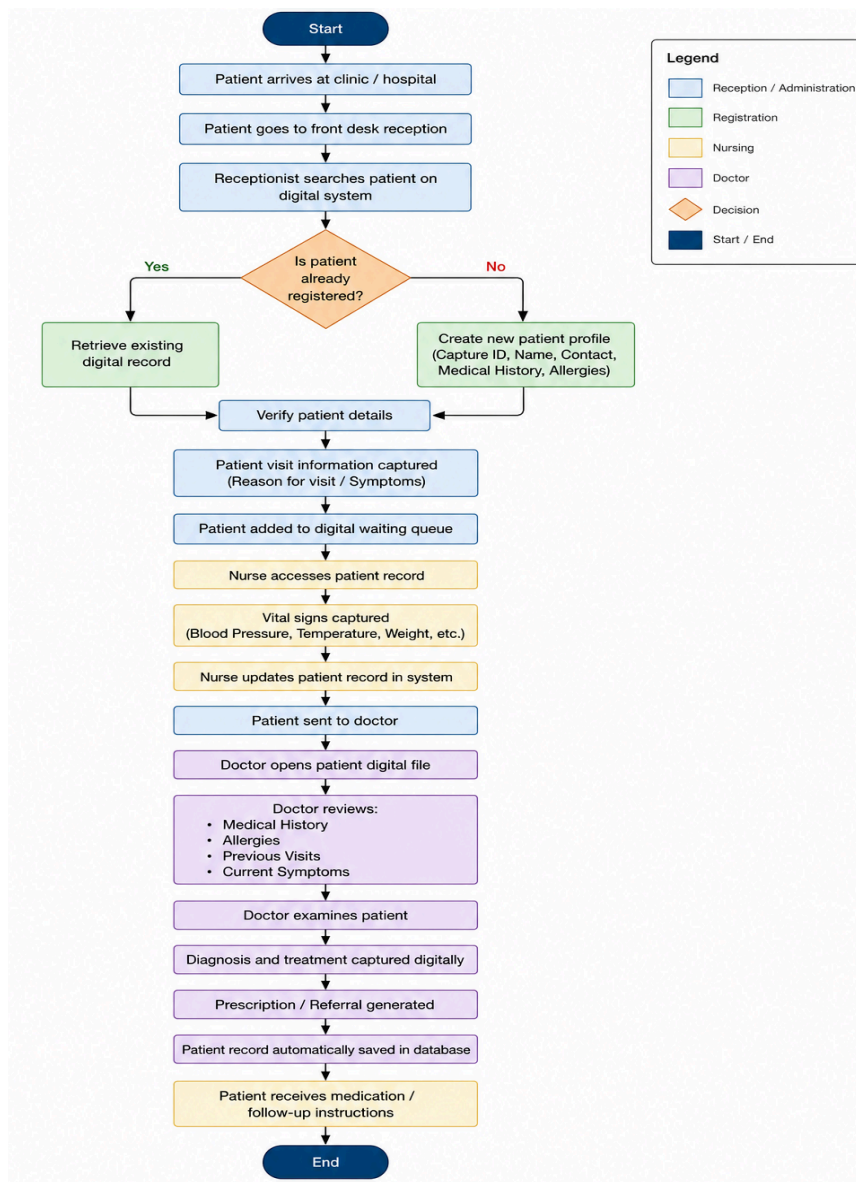
5. User Requirements

- Receptionist must be able to register and search patients
- Doctor must be able to view and update patient medical records
- Administrator must be able to manage users and maintain the system

6. GUI - Patient data fields that can be collected and stored

- Patient ID
- First name
- Surname
- Date of birth
- Gender
- Phone number
- Home address
- Emergency contact
- Medical history
- Consultation date
- Diagnosis
- Treatment given

3. Process flowchart



4. SQL Data Collection Plan

Data to be collected	Source	Responsible Person	SQL table
Patient personal details	Front desk/ patient ID	Receptionist	Patients
Contact details	Patient form	Receptionist	Patients
Medical history	Existing paper files	Data capturer/ nurse	Records
Allergies	Patient file/ patient interview	Nurse	Records
Visit information	Current consultation	Nurse / doctor	Visits
Diagnosis	Doctor consultation	Doctor	Visits
Treatment/ prescription	Doctor notes	Doctor	Visits

5. Migration Checklist

- ☐ Identify all paper-based patient files
- ☐ Sort files according to patient number, ID, or surname
- ☐ Check files for missing or unclear information
- ☐ Prepare standard data capturing form
- ☐ Create SQL database and required tables
- ☐ Capture patient information in the database
- ☐ Validate captured data for accuracy
- ☐ Remove duplicate patient records
- ☐ Test database search and retrieval
- ☐ Backup the database
- ☐ Train clinic staff on how to use the system
- ☐ Approve the system for live use
- ☐ Start using the digital system